

C++ Practice Questions

CSE141 Introduction to Programming

Instructions

- Write a separate C++ program for each question unless your instructor says otherwise.
- Use loops and arrays directly; avoid STL containers and ready-made sorting/searching algorithms unless a question explicitly permits them.
- For array questions, assume a reasonable maximum size and validate the logical size entered by the user.
- For matrix questions, pay close attention to row/column boundaries so that no cell is printed or counted accidentally twice.
- For searching and sorting questions, implement the stated algorithm yourself rather than calling a library function.
- Questions are divided into **Easy**, **Medium**, and **Hard**. Some later questions intentionally combine several topics.

Topic 1: For Loops

Question 1 **[Easy]**.....

Print all integers from 1 to N, where N is entered by the user. On the next line, print the same values in reverse order. Use for loops for both traversals.

Question 2 **[Easy]**.....

Read an integer N and display the first N multiples of 7 in the form $7 \times i = \text{result}$. Also calculate and display the sum of these multiples.

Question 3 **[Medium]**.....

Read two integers start and end. Print every number in this inclusive range that is divisible by 3 but not divisible by 5, then display how many such numbers were found.

Question 4 **[Medium]**.....

Read a positive integer and calculate its factorial using a for loop. Show the multiplication sequence as well as the final factorial value.

Question 5 **[Medium]**.....

Read an integer N and print the first N terms of the Fibonacci sequence. Assume the sequence begins with 0 and 1.

Question 6 **[Hard]**.....

Read a positive integer N and determine whether it is prime by testing possible divisors with a for loop. Stop unnecessary checking once a divisor is found.

Question 7 **[Hard]**.....

Read two positive integers A and B and find their greatest common divisor by checking candidate divisors with a loop. Display the largest common divisor found.

Question 8 **[Hard]**.....

Read N and print all prime numbers from 2 through N. For each candidate, use an inner checking process rather than hard-coding known primes.

Topic 2: Nested For Loops and Patterns

Question 9 **[Easy]**.....

Read rows and columns and print a rectangular grid of asterisks with exactly the requested dimensions. Use one loop for rows and one loop for columns.

Question 10 **[Easy]**.....

Read N and print an N by N multiplication table in grid form. Row i and column j should contain $i*j$.

Question 11 **[Medium]**.....

Read N and print a right-aligned triangle of numbers. Row r should contain the values 1 through r, with spacing chosen so the shape remains aligned.

Question 12 **[Medium]**.....

Read N and print a hollow N by N square using asterisks on the boundary and spaces inside. Handle small values of N correctly.

Question 13 **[Medium]**.....

Read N and print a checkerboard of 0 and 1 values. Adjacent cells horizontally and vertically must alternate.

Question 14 **[Hard]**.....

Read an odd positive N and print an X pattern in an N by N character grid. Print an asterisk where the row and column lie on either diagonal and a dot elsewhere.

Question 15 **[Hard]**.....

Read N and print a concentric number-square pattern. For $N=4$, the outside layer contains 4, the next layer 3, and the center decreases toward 1.

Question 16 **[Hard]**.....

Read N and generate a triangular pattern in which each position contains the sum of the two positions above it, starting with 1 at the top. You may use a 2D array to store previously calculated values.

Topic 3: While and Do-While Loops

Question 17 *[Easy]*.....

Repeatedly read integers using a while loop until the user enters 0. Display the sum and count of all non-zero values entered.

Question 18 *[Easy]*.....

Read a positive integer and use a while loop to count how many digits it contains. Treat 0 as a one-digit number.

Question 19 *[Medium]*.....

Read an integer and reverse its digits using repeated division and remainder operations. For example, 4821 should produce 1284.

Question 20 *[Medium]*.....

Use a do-while loop to create a menu with options to add, subtract, multiply, or exit. The menu must appear at least once and repeat until Exit is selected.

Question 21 *[Medium]*.....

Keep asking for a password using a do-while loop until the correct password is entered or three attempts have been used. Display an appropriate final message.

Question 22 *[Hard]*.....

Read a positive integer and repeatedly replace it by the sum of its digits until only one digit remains. Display each intermediate value.

Question 23 *[Hard]*.....

Create a number-guessing process with a fixed secret number. Use a do-while loop to keep accepting guesses and report Too High or Too Low until the correct value is entered.

Topic 4: Type Casting: Implicit and Explicit

Question 24 *[Easy]*.....

Store an int and a double, add them, and display the result. State through program output which value is implicitly converted during the expression.

Question 25 *[Easy]*.....

Read two integers representing obtained marks and total marks. Use explicit casting so the calculated percentage keeps its decimal part.

Question 26 *[Medium]*.....

Read a double amount such as 127.85. Explicitly cast it to int to obtain the whole-number portion, then calculate the remaining fractional portion.

Question 27 *[Medium]*.....

Read a character and display its integer character code using explicit casting. Then read an integer code and display the corresponding character.

Question 28 *[Medium]*.....

Given total seconds as an integer, calculate hours, minutes, and seconds. Where a decimal average is later required, use casting deliberately rather than changing all variables to double.

Question 29 *[Hard]*.....

Read three integer test scores. Display both the integer average produced by integer division and the precise decimal average produced using explicit casting. Explain the difference through labels in the output.

Question 30 *[Hard]*.....

Read a floating-point temperature and explicitly convert it to int before and after adding 0.5. Compare the results to demonstrate truncation; do not use a built-in rounding function.

Topic 5: One-Dimensional Arrays

Question 31 *[Easy]*.....

Read 10 integers into an array, then print them in the original order and in reverse order without modifying the stored array.

Question 32 *[Easy]*.....

Read N values into an array and calculate the sum, arithmetic mean, minimum value, and maximum value in one traversal after input.

Question 33 *[Medium]*.....

Read N integers and count how many are positive, negative, zero, even, and odd. A value may contribute to more than one relevant category.

Question 34 *[Medium]*.....

Read an array and create a second array containing the elements in reverse order. Print both arrays so the mapping can be checked.

Question 35 *[Medium]*.....

Read N integers and shift every element one position to the right, moving the last element to index 0. Perform a circular rotation by one position.

Question 36 *[Hard]*.....

Read N integers and remove duplicate values logically by building a second array containing only the first occurrence of each value. Do not use sets or library containers.

Question 37 *[Hard]*.....

Read two arrays of equal size and construct a third array by alternating elements: first from A, first from B, second from A, second from B, and so on.

Question 38 *[Hard]*.....

Read an array and find the second-largest distinct value without sorting the array. Clearly handle the case where no second distinct value exists.

Topic 6: Two-Dimensional Arrays and Calculations

Question 39 *[Easy]*.....

Read a 3 by 4 integer matrix row by row and print it in rectangular form. Then display the sum of each row.

Question 40 *[Easy]*.....

Read a 4 by 4 matrix and display the sum of each column. Afterward, identify the column having the largest sum.

Question 41 *[Medium]*.....

Read an N by N matrix and calculate the sums of the main diagonal and secondary diagonal. If N is odd, make sure the center element is handled correctly when calculating a combined diagonal sum.

Question 42 *[Medium]*.....

Read two matrices of the same dimensions and create a third matrix containing their element-wise sum. Print all three matrices clearly.

Question 43 *[Medium]*.....

Read an N by M matrix and find its largest and smallest values together with the row and column index of each.

Question 44 *[Hard]*.....

Read an N by N matrix and determine whether it is symmetric about the main diagonal, meaning $\text{grid}[i][j]$ equals $\text{grid}[j][i]$ for every valid pair.

Question 45 *[Hard]*.....

Read an N by M matrix and create its transpose in a separate M by N matrix. Print the original and transposed matrices.

Question 46 *[Hard]*.....

Read an N by N matrix and calculate the sum of the border cells only. Corner cells must be counted exactly once.

Topic 7: Matrix Traversal Variations

Question 47 *[Easy]*.....

Read an R by C grid row by row, then print it column by column: finish an entire column before moving to the next column.

Question 48 *[Medium]*.....

Read an R by C grid and print it in snake order: first row left-to-right, second row right-to-left, third row left-to-right, and so on.

Question 49 *[Medium]*.....

Read an N by N grid and print only the cells on its outer boundary in clockwise order, beginning at the top-left corner. Do not print any corner twice.

Question 50 *[Hard]*.....

Read an R by C grid and print every value in a clockwise spiral beginning at the top-left corner and moving right first. Continue inward until all values are printed exactly once.

Question 51 *[Hard]*.....

Read an R by C grid and print every value in a clockwise spiral beginning at the bottom-left corner and moving upward first. Continue shrinking the unvisited boundaries.

Question 52 *[Hard]*.....

Read an N by N grid and print its values from the center outward in a spiral. For odd N begin at the single center cell; move right first, then continue in an expanding spiral until every cell is printed.

Question 53 *[Hard]*.....

Read an N by N grid and print its values diagonally by increasing row+column index: first grid[0][0], then the next anti-diagonal, continuing until the bottom-right cell.

Question 54 *[Hard]*.....

Read an N by N grid and rotate it 90 degrees clockwise into a second matrix. Do not use a built-in matrix or rotation function.

Topic 8: Linear and Binary Search

Question 55 *[Easy]*.....

Read N integers and a target value. Use linear search to report the index of the first occurrence of the target, or -1 if it is absent.

Question 56 *[Medium]*.....

Read N integers and a target. Use linear search to print every index at which the target occurs and the total number of occurrences.

Question 57 *[Medium]*.....

Read a sorted array and a target. Implement binary search using low, high, and mid indices and report whether the target exists.

Question 58 *[Hard]*.....

Read a sorted array that may contain duplicates. Use a binary-search-based approach to find the first occurrence of a target rather than stopping at an arbitrary matching index.

Question 59 *[Hard]*.....

Read a sorted array and a target that may not be present. Use binary search logic to find the position where the target should be inserted so that the array would remain sorted.

Question 60 *[Hard]*.....

Given a sorted array, use binary search to determine whether a target exists, while printing the low, mid, and high index used at every iteration so the narrowing search interval can be traced.

Topic 9: Bubble Sort and Selection Sort

Question 61 **[Easy]**.....

Read N integers and sort them in ascending order using bubble sort. After each complete pass, print the current state of the array.

Question 62 **[Medium]**.....

Modify bubble sort so it stops early when a complete pass performs no swaps. Display the number of passes actually executed.

Question 63 **[Medium]**.....

Read N integers and sort them in descending order using bubble sort. Count and display the total number of swaps.

Question 64 **[Easy]**.....

Read N integers and sort them in ascending order using selection sort. After each pass, print the selected minimum value and the array state.

Question 65 **[Medium]**.....

Use selection sort to arrange values in descending order by selecting the largest remaining value for each position.

Question 66 **[Hard]**.....

Read the same input array into two copies. Sort one using bubble sort and the other using selection sort. Count comparisons and swaps for each algorithm and display the counts; do not claim one is universally faster from a single test.

Question 67 **[Hard]**.....

Read N student marks into an array, sort them from highest to lowest using either bubble or selection sort, then display the highest, lowest, median position(s), and sorted marks.

Topic 10: Mixed Loop and Array Challenges

Question 68 **[Medium]**

Read N integers. For each element, count how many elements to its right are smaller than it, and store that count in a second array.

Question 69 **[Medium]**

Read an array and determine whether it is already sorted in ascending order, descending order, or neither. Equal neighboring values are allowed.

Question 70 **[Hard]**

Read N integers and move all zero values to the end while preserving the relative order of all non-zero values. Use arrays and loops only.

Question 71 **[Hard]**

Read an N by N matrix and find the row with the greatest sum and the column with the greatest sum. Display both indices and sums.

Question 72 **[Hard]**

Read an N by N matrix and determine whether every row and every column has the same sum. Also calculate both diagonal sums and report whether all these sums are equal.

Question 73 **[Hard]**

Read a grid containing only 0 and 1. Count the number of horizontal adjacent pairs of 1s and vertical adjacent pairs of 1s. Do not count diagonal neighbors.

Question 74 **[Hard]**

Read an N by N matrix and search for a target using a normal nested-loop scan. Report every coordinate where it appears, then display the total number of matches.

Question 75 **[Hard]**

Read an array, sort it using selection sort, then ask for a target and search the sorted result using binary search. Print the sorted array before performing the search.

Topic 11: Optional Challenge Problems

Question 76

[Hard].....

Spiral reconstruction: read $R \times C$ values as a one-dimensional sequence and place them into an R by C matrix in clockwise spiral order starting from the top-left. Print the completed matrix.

Question 77

[Hard].....

Ring sums: for an N by N matrix, calculate the sum of each concentric rectangular layer separately, starting with the outermost layer and moving inward.

Question 78

[Hard].....

Saddle point: find every matrix element that is the smallest value in its row but the largest value in its column. Print each value and its coordinates.

Question 79

[Hard].....

Array equilibrium: find every index where the sum of elements strictly to its left equals the sum of elements strictly to its right. Avoid recalculating both full sums from scratch for every index.

Question 80

[Hard].....

Binary-search guessing: for a sorted array, simulate the indices visited by binary search for a target and print the path of inspected values until the target is found or the interval becomes empty.

Question 81

[Hard].....

Sort verification: implement both bubble sort and selection sort on separate copies of the same array, then use a loop to verify that both resulting arrays are identical.